

Home Field Scope

A short guide for the 7-inch tracker

What this is

Home Field Scope is a small screen that sits on a table and shows what is going on in the sky around your airport. Planes in range, the weather at your field, the time, tonight's sky, rocket launches, and the space station.

It does all of that over your home Wi-Fi. It does not have its own radio, and it does not need a window or an antenna. If the Wi-Fi is down, the screen can still be used, but the live lists stay empty until the connection comes back. Photos in the screen saver are the one thing that does not need Wi-Fi. Those come from a card you put in the tracker.

Getting started

Plug the USB cable into power. The screen lights up. The first time you turn it on, it opens a page called JOIN WIFI,

Tap SCAN. Your nearby networks show up, two at a time. The arrows on the left and right move through the list. Tap the name of your network. It fills in the name for you.

Tap the password line and type the password on the keyboard.

abc switches between small letters and capitals. !# brings up numbers and symbols. Tap it again to get letters back. SPACE adds a space. DEL removes the last letter. SAVE stores the name and password, then the tracker restarts and joins that network. You will see the words saved, then linking. RESET WIFI forgets the network name and password. Your other settings stay. FACTORY forgets everything and starts over. Use that only when you want a clean start. More on that later.

If the scan finds nothing, you can still tap the name line and type the network yourself, then type the password and tap SAVE.

After it joins, you land on the home screen. Out of the box that screen is centered on Fresno, California. Change it to your own airport before you settle in. Tap SETUP along the bottom, stay on LOCATION, and follow the steps in the Setup section below.

Give it a little while after plugging in. The corner of the home screen says linking, then MAP, then the traffic starts to fill in. A full minute is normal the first time.

How to get around

Along the bottom of the home screen are five buttons: WX, LIST, FIELD, SKY, and SETUP.

Almost every other page has BACK at the top left and HOME at the top right.

BACK returns to the page you just came from. From the clock, the launches, the night sky, or the space station, BACK returns you to the Sky page when that is where you came from.

HOME always takes you back to the radar.

Tap one thing, wait for the page to finish drawing, then tap the next. Moving through pages too quickly can freeze the tracker, or leave stray lines across the home map. If that happens, unplug the USB cable, wait a few seconds, and plug it back in. Your Wi-Fi name, your airport, and your settings are still saved. You only lose whatever was on the screen at that moment.

The home screen

This is the radar. Your airport is the dot in the middle. Rings around it are distance, in nautical miles. A nautical mile is a little longer than a regular mile. North is up.

Each plane is a dot with a short line. The line is the direction that plane is flying. The name sits in a small box next to the dot.

The top left corner tells you the live status. A number of seconds is how long ago the traffic was refreshed, followed by how many planes are in range. MAP means the map picture is still downloading. TRAFFIC means it is still fetching planes. NO WIFI means it is not on the network. none in range means the connection is fine and the sky around you is simply empty.

The top right is the clock, once the time has come in over Wi-Fi. The Wi-Fi mark sits at the far right. Arcs mean connected. A slash through it means it is not.

ZOOM - and ZOOM + sit just above the bottom buttons. They change how much sky you are looking at, from 5 nautical miles out to 60. The number in the middle is the range you are on now. You can also pinch with two fingers on the map. Spreading your fingers zooms in. Pinching them together zooms out. A wider range shows more planes, and they sit closer together on the screen.

Tap a plane, or tap its name box, to open that flight.

What the names mean

The name on a plane is whatever that aircraft is sending.

An airline flight looks like letters followed by numbers. SWA1842 is a Southwest flight. AAL is American, UAL is United, DAL is Delta, and so on. The letters are the airline. The numbers are that flight.

If the plane is not on an airline flight, the tracker shows the tail number instead. In the United States that usually starts with N, like N172SP. That is the registration painted on the aircraft, not a flight you can book.

A row of dashes means the plane did not send a name. You can still see it on the map, and you can still open it, but there is often nothing more to look up.

Why the colors are different

The color is the altitude, not the airline.

Amber is a plane below 5,000 feet. Those are the ones climbing out, coming in to land, or flying low.

White is a plane from 5,000 feet up to just under 20,000.

Blue is a plane at 20,000 feet or higher. Those are usually the ones passing overhead on a longer trip.

The same color is used on the map and in the traffic list, so a blue name in the list is a high aircraft.

A magenta ring around a dot means that is the plane you have open. The magenta dot in the center of the map is your airport, not a plane.

Why some flights have details and some do not

Tap a plane and a details page opens. The big name at the top is the call sign or tail number. Under that you may see the maker, the kind of airplane, and the registration.

When the tracker can find the trip, it also shows the airline, where it departed, and where it is landing. It shows the off time and the landing time when those were published, how long the trip is, the altitude, the speed in knots and in miles per hour, the heading, a top speed if one has been seen, and the four-digit code air traffic control assigned. Seats, a cruise speed, and an owner show up only when that information was found.

Airline flights usually fill in. Give the page a few seconds. The lookup happens over Wi-Fi after you tap the plane.

Not every flight gets a city. The tracker writes a departure and a landing only when that trip passes near the airplane. A flight number can have another trip saved under it, sometimes far from where the airplane actually is. When that happens the city line stays blank. The

airplane stays on the map. That is normal.

Some aircraft will open with a lot of dashes. That is normal. Private planes, military planes, and planes that are not broadcasting a flight number often have no published route. A name made of dashes usually cannot be looked up at all. You will still see the altitude and speed from the live signal. The cities and the times stay blank.

Sometimes the departure or the landing shows an airport code instead of a city name. You might see KLAX, SFO, KFAT, or something like that, rather than Los Angeles or San Francisco. That means the tracker found the airport, and it does not have a city name stored for that one. The code is still the right airport. A line of dashes means it did not find a departure or a destination at all.

If the page says no aircraft, that plane has already left your range.

A plane that does not belong

The tracker does not have its own radio. Aircraft send out their position and flight number. Those signals are called ADS-B. A service on the internet collects them and passes them to the tracker. The tracker shows what it was sent. A position, a flight number, or a city in that information can be wrong, so the screen can make a mistake. It does not happen on every flight, but it is possible. Leave the map up and the next refresh often drops a bad dot. A blank city line means the tracker would not attach a trip that does not pass near that airplane.

Weather

WX is the weather at your home airport, drawn over the same map.

The top line is your airport code, the temperature, and a short word for the sky: clear, partly cloudy, cloudy, fog, rain, snow, thunder, windy, or tornado. Under the map you get the wind direction and speed, the humidity, and the air quality with a plain word such as good or moderate. The small line at the very bottom is the weather report as it arrived. It can look terse. That is the original report, not a summary.

If it says waiting for live weather, it is still fetching. Weather refreshes on its own while you leave this page up.

Traffic list

LIST is every plane currently in range, in a table.

CALL is the flight or tail number. TYPE is a short code for the kind of airplane, such as a 737 or a small Cessna. ALT is altitude in feet. KT is speed in knots. MPH is that same speed in

miles per hour.

The row is colored the same way as the dot on the map. Tap a row to open that flight.

Twelve planes fit on the page. If there are more, tap the bottom of the list, where it says tap a row, to bring up the next set. Tap it again after the last set and it returns to the top. Do this slowly. Tapping through the pages in a rush is one of the things that can freeze the screen or leave lines on the home map.

If it says none in range, nothing is inside the circle you have set.

Field

FIELD is your airport, written out.

The big code on the left is your home field, with the city under it when the tracker knows that airport. On the right is the nearest plane, with its distance in nautical miles and its altitude.

Under that, TWR is the tower frequency and ATIS is the recorded weather broadcast, for airports the tracker already knows. A custom spot you typed in by latitude will show dashes there, because there is no tower frequency to look up.

The table under that is the planes again, nearest first.

CALL and TYPE are the same as on the list. ALT is altitude. NM is how far the plane is from your airport, in nautical miles. BRG is the compass direction from you to that plane. 000 is north, 090 is east, 180 is south, 270 is west.

Sky

SKY is four large buttons.

SKY/CLOCK is the clock and the local weather. TODAY'S LAUNCHES is the next rocket launches. NIGHT SKY is what you can see after dark. ISS is the space station.

The Sky page itself opens right away. The clock and the weather can take about a minute the first time you open them. Once they have arrived, you can leave and come back without waiting through that first minute again. They load again after you unplug the power. The world picture on the space station page is already on the tracker.

Clock

The large time is local time for your home airport, not a generic clock. SUN UP and SUN DOWN are today's sunrise and sunset there.

Beside the weather icon is the temperature. Under that is the high and the low, and either umbrella or dry, which is the tracker's way of saying rain is likely or it is not. AIR is the air

temperature, feels is what it feels like, and the rest is humidity and wind. At night a clear sky shows the moon instead of the sun when the moon is actually up.

If it says waiting for live clock, the time has not come in yet. Leave the page up. It redraws on its own.

Today's launches

Up to four upcoming launches. Each card has the day, the place, the time, and who is flying it. If the live list has nothing coming up, the page says so. If the clock is not ready yet, this page waits too.

Night sky

The top line is the moon, as a percent of full, when the time is known. Under that is a word for the sky at your airport, the same kind of word as on the weather page.

Then a short list of what is up: the name, the direction to look (N, NE, E, and so on), how high it is in degrees, and the hours it is up. Higher on that list is the easier thing to find. If nothing is high enough, it says none up tonight. The list follows your home airport, so changing the airport changes the sky.

Space station

A world map. The bright side is day, the dim side is night, and the small sun marks where the sun is overhead. Your airport is the magenta dot. The station is the small mark moving along a blue path.

When the station is overhead, or when a pass starts within about twenty minutes, a banner appears that says LOOK. It tells you which way to face and how high to look, or it counts down the minutes until the pass. Step outside and use that direction.

Under the map is VISIBLE PASSES. These are passes you can actually see: the station is high enough, and the sky is dark enough. Each line is the date, the local time, the direction it comes from, and how long it stays up. The tracker looks ahead ten days. checking passes means it is still working that out. none in the next 10 days means there is not a visible pass in that window. waiting for live ISS data, or waiting for live clock, means the Wi-Fi or the time is not ready yet.

The world picture is already on the tracker. The path and the list of passes fill in once it is on Wi-Fi.

Setup

Tap SETUP on the home screen. You get three tabs: DISPLAY, LOCATION, and ABOUT. The home button opens LOCATION, which is the one to set first.

Location

AIRPORT CODE is four letters. Tap a letter to step it to the next letter. If the four letters match an airport the tracker knows, the city and the position fill in by themselves.

The arrows under the code step through the airports already stored in the tracker. The city name in the middle changes as you go. Move through them slowly. Stepping through locations too fast can mess up the screen. The processor in this tracker is not that powerful. Wait until the city name settles before you tap the arrow again. custom means the code is not one of those stored airports. You can still use it. Type the position yourself.

TYPE LAT opens a keypad for the latitude. TYPE LON does the same for the longitude. N and S nudge the latitude. W and E nudge the longitude. North is a positive number. South is negative. West is negative. East is positive. The example on the keypad shows the pattern.

The line under the numbers is there as a reminder. Planes, the sky, the weather, and the space station all use this spot. The clock's time zone follows it too.

Tap APPLY FIELD when the airport looks right. The tracker saves it and starts over with planes, weather, and passes for that place.

Display

RANGE is the same circle as the zoom on the home screen. Minus and plus move it in steps of 5 nautical miles, from 5 out to 60.

STREET is an ordinary map. SAT is a satellite picture. OFF leaves the radar on a plain background, which is easier to read in some rooms. SATELLITE and MAP, lower down, do the same job as SAT and STREET.

DARK and DAY change the look of the screen. Day is the light background. Dark is for a dim room. Changing this also reloads the street map so the picture matches.

LIGHT is the brightness: LOW, MED, or HIGH.

F shows temperatures in Fahrenheit. C shows them in Celsius. Altitude stays in feet either way. Speeds stay in knots and miles per hour.

SCREEN SAVER is OFF, 2m, 5m, or 15m. That is how long the tracker waits, with nobody touching it, before the photos start. WIFI on this row opens the join page, the same one you

used at the start, if you need to change networks later.

About

This page shows the firmware version, the number of the software on the tracker.

CHECK looks online, over Wi-Fi, for a newer version. If you already have the newest, it says so. If a newer one exists, it tells you to tap INSTALL.

INSTALL downloads that newer version and restarts the tracker. Leave the USB plugged in the whole time. Do not pull power while it says it is downloading.

While it updates, the picture can jump, flash, or look scrambled. That is normal. After the update the tracker may restart itself several times. Leave it alone and let it finish. If it is still playing up one minute after those restarts begin, unplug the USB cable and plug it back in.

FROM SD installs a file from the card instead of from the internet. The file on the card should be named HFS.BIN. The names hfs.bin, hfs_7in.bin, and firmware.bin are also accepted. Put the file on the card, tap FROM SD, and leave it plugged in until it restarts.

If INSTALL says it cannot update, use FROM SD, or get in touch with the seller. Do not keep tapping INSTALL.

Screen saver

Put a memory card in the tracker. The card should be formatted FAT32. Copy your photos onto it as JPG pictures. The tracker looks for files ending in .jpg or .jpeg, and it can hold a list of up to 24 of them. Skip files whose names start with a dot.

In SETUP, on the DISPLAY tab, pick 2m, 5m, or 15m. After the screen has been left alone for that long, it goes black and starts showing the photos, about ten seconds each, then the next one.

Touch anywhere to leave. You return to the page you were on.

The saver does not start while you are on the Wi-Fi page, so you can take your time typing a password.

If there is no card, the screen says so and asks for a FAT32 card. If the card is in but no photos were found, it says that too. The saver is only photos. It does not turn the tracker off, and it does not stop the tracker from using Wi-Fi once you wake it.

Reset

There are two different resets, and both of them live on the JOIN WIFI page. From the home screen, tap SETUP, then WIFI at the bottom of the DISPLAY tab.

RESET WIFI clears the network name and the password. The airport, the range, the map style, and the screen saver stay as they were. You will need to join a network again.

FACTORY clears all of it. Wi-Fi, airport, range, map, day or dark, brightness, Fahrenheit or Celsius, and the screen saver all go back to the way the tracker arrived. It restarts on its own, opens JOIN WIFI, and the home airport goes back to Fresno until you change it. Use FACTORY when you are giving the tracker to someone else, or when you want to set it up from the beginning.

If the screen misbehaves

Move slowly between pages. A pause of a second between taps is enough.

If the screen plays up at any time, unplug the USB cable, wait a few seconds, and plug it back in. That is a full restart. Your saved Wi-Fi and your saved airport come back with it. Lines across the home map, a frozen page, or a scrambled picture are the same fix.

If the corner says NO WIFI, the network dropped or the password was wrong. Tap SETUP, then WIFI, and join again. A wrong password still lets you onto the home screen. It just cannot load anything.

If a page says it is waiting, leave it up. The clock, the weather, the launches, and the space station all arrive on their own once Wi-Fi is up. The first visit takes the longest.

If a flight's cities never fill in, the tracker did not have a trip that passes near that airplane. The plane is still real. Try another airliner if you want to see a full details page.

Pull the memory card only when the screen saver is not running, and not while FROM SD is installing.

A few things worth knowing

Everything live comes through Wi-Fi only. Planes, the map picture, weather, the clock, launches, the night sky, and the space station. There is nothing to point at the sky.

The map is a downloaded picture of the ground under your airport. It is not a live camera.

The tracker remembers your settings when you unplug it. It does not remember the planes. Those load again each time it starts.

Altitude is in feet. Distance is in nautical miles. Speed is shown both in knots, which is how aviation measures it, and in miles per hour.

The list and the field page only include planes inside the range you set. Zoom out if the sky looks empty and you expected more.

Weather, sunrise, the night sky, and the space station passes all follow the airport you set,

including the local time. Set the airport before you trust the clock.

A visible space station pass can be days away. none in the next 10 days is a real answer, not an error.

The screen is happy to stay on. Leave it plugged in and it keeps refreshing. The screen saver is there for when you want your own photos instead of the radar.